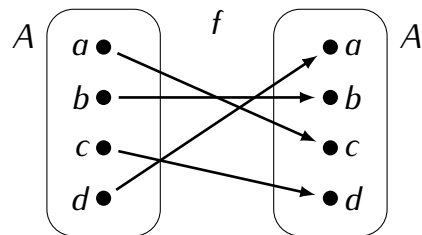


**Definition 10.1**

A *permutation* of a set  $A$  is a function  $f: A \rightarrow A$  which is a bijection.

**Definition 10.2**

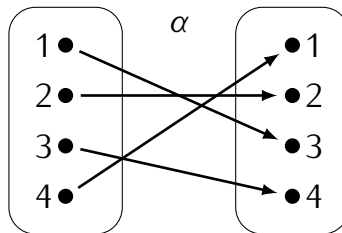
Let  $A$  be a set. The *permutation group* of  $A$  is a group  $S(A)$  defined as follows:

- **Elements of  $S(A)$ :** permutations  $f: A \rightarrow A$ .
- **Group operation:** composition of functions  $g \circ f$ .
- **The identity element:** the function  $\varepsilon: A \rightarrow A$ ,  $\varepsilon(x) = x$  for all  $x \in A$ .
- **The inverse of  $f$ :** the inverse permutation  $f^{-1}$ .

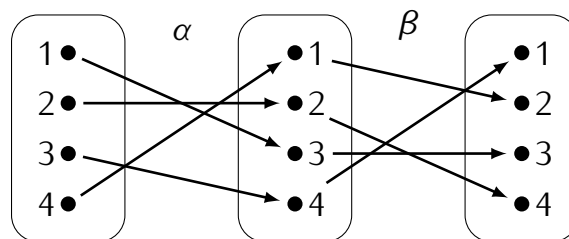
### Definition 10.3

For  $n \geq 1$  the group  $S_n$  is the group of permutations of the set  $A = \{1, 2, \dots, n\}$ . This group is called the *symmetric group on  $n$  letters*.

Matrix notation of permutations:



Composition:



### Theorem 10.4

For any  $n \geq 1$  we have  $|S_n| = n!$

## Dihedral groups and permutation groups

**Note.** The groups  $S_n$  are non-abelian for  $n > 2$

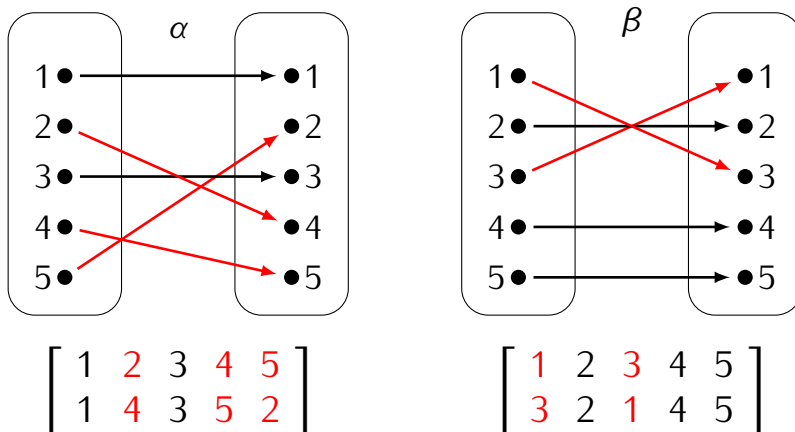
### Definition 10.5

Let  $\alpha \in S_n$  and let  $i \in \{1, \dots, n\}$ . We will say that  $\alpha$  *moves*  $i$  if  $\alpha(i) \neq i$ . If  $\alpha(i) = i$  we will say that  $\alpha$  *fixes*  $i$ .

### Definition 10.6

We will say that permutations  $\alpha, \beta \in S_n$  are *disjoint* if there is no  $i \in \{1, \dots, n\}$  which is moved by both  $\alpha$  and  $\beta$ .

**Example.**



### Theorem 10.7

If  $\alpha, \beta \in S_n$  are disjoint permutations then

$$\alpha \circ \beta = \beta \circ \alpha$$

Moreover,

$$\alpha \circ \beta(i) = \begin{cases} \alpha(i) & \text{if } i \text{ is moved by } \alpha \\ \beta(i) & \text{if } i \text{ is moved by } \beta \\ i & \text{otherwise} \end{cases}$$

**Definition 10.8**

A permutation  $\alpha \in S_n$  is a *cycle of length  $r$*  (or  *$r$ -cycle*) if there are distinct elements  $i_1, i_2, \dots, i_r \in \{1, 2, \dots, n\}$  such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and  $\alpha$  fixes all other elements of  $\{1, \dots, n\}$ .

**Theorem 10.9**

Every permutation in  $S_n$  is either a cycle or a product of disjoint cycles.

### Lemma 10.10

Let  $\alpha \in S_n$ , and let  $i_0 \in \{1, \dots, n\}$  be an element moved by  $\alpha$ . Then:

- 1) There exists  $r > 1$  such that  $\alpha^r(i_0) = i_0$
- 2) If  $r > 1$  is the smallest integer satisfying  $\alpha^r(i_0) = i_0$  then all elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are distinct.

*Poof of Theorem 10.9.*

Recall that the least common multiple of integers  $n_1, n_2, \dots, n_k \geq 1$  is the smallest positive integer  $\text{lcm}(n_1, \dots, n_k)$  which is divisible by each of these numbers.

**Theorem 10.11**

Assume that a permutation  $\alpha \in S_n$  has a decomposition into disjoint cycles

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where  $\gamma_i$  is a cycle of length  $r_i > 1$ . Then the order of  $\alpha$  is given by

$$|\alpha| = \text{lcm}(r_1, r_2, \dots, r_m)$$

**Exercise.** Compute the order of the following permutation in  $S_8$ :

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix}$$

**Exercise.** Find all possible orders of elements of  $S_5$ .

**Exercise.** Compute the number of permutations of order 10 in  $S_8$ .

**Exercise.** Compute the number of permutations of order 3 in  $S_7$ .

**Definition 10.12**

A *transposition* in  $S_n$  is a cycle  $(i_1, i_2)$  of length 2.

**Theorem 10.13**

Every permutation in  $S_n$  can be written as a product of transpositions.

### Theorem 10.14

Let  $\alpha \in S_n$  and let

$$\alpha = \beta_1 \circ \beta_2 \circ \cdots \circ \beta_r$$

be a decomposition of  $\alpha$  into a product of transpositions.

- If the number  $r$  is even, then every other decomposition of  $\alpha$  into transpositions consists of an even number of transpositions.
- If  $r$  is odd, then every other decomposition of  $\alpha$  into transpositions consists of an odd number of transpositions.

**Lemma 10.15**

Let  $\beta_1, \dots, \beta_r$  be transpositions in  $S_n$  such that

$$\beta_1 \circ \beta_2 \circ \cdots \circ \beta_r = \varepsilon$$

where  $\varepsilon$  is the identity permutation. Then  $r$  is an even number.

**Definition 10.16**

A permutation  $\alpha \in S_n$  is *even* if it can be written as a product of even number of transpositions and it is *odd* if it can be written as a product of an odd number of transpositions.

**Theorem 10.17**

The subset of  $S_n$  consisting of all even permutations is a subgroup of  $S_n$ .

**Definition 10.18**

The subgroup of  $S_n$  consisting of even permutations is called an *alternating group on  $n$  letters* and it is denoted by  $A_n$

**Theorem 10.19**

For  $n \geq 2$  the alternating group  $A_n$  has order  $\frac{n!}{2}$ .

**Definition 10.20**

The *sign* of a permutation  $\alpha \in S_n$  is defined as follows:

$$\text{sign}(\alpha) = \begin{cases} +1 & \text{if } \alpha \text{ is even} \\ -1 & \text{if } \alpha \text{ is odd} \end{cases}$$